On Growth and Form of Wildfire

NSF FIRE-MODEL Proposal Narrative

Project Summary (Target: 750 words total)

1) Overview

Target length: 300 words. Wildfire models are most likely to fail in the conditions where their predictions matter most. Large fires do not grow as steady extensions of local spread; they reorganize. Periods of patchy, ignition-limited growth give way to rapid acceleration, directional alignment, and the emergence of coherent fronts whose behavior is no longer well described by local assumptions. Yet most existing models treat wildfire growth as governed by a single process across all times and conditions (Rothermel, 1972; Finney, 1998). This mismatch between assumption and reality limits both explanation and prediction precisely in high-impact events.

This reorganization is visible in the fire itself. Early growth is irregular, fragmented, and strongly influenced by local conditions. As continuity emerges, the fire takes on a coherent form, and its behavior becomes increasingly constrained by the structure of the advancing boundary. What changes is not only how fast the fire spreads, but what determines that spread.

This mismatch is most consequential under the conditions where prediction is most difficult and most important. When fires transition between modes of growth, small differences in forcing can produce large differences in outcome. Models that do not represent these transitions explicitly are least reliable precisely when the system changes most rapidly (Sullivan, 2009; Finney et al., 2013).

What follows from this is not subtle. Boundaries that were once irregular and disconnected become continuous and directional (Mandelbrot, 1983). Growth accelerates, not because each point spreads faster, but because the system has reorganized into a form that sustains that growth. What changes is not only the rate of spread, but the structure that determines it.

Wildfire growth is treated here not as a single process, but as a sequence of regimes with distinct controlling mechanisms. In early stages, spread is fragmented and locally constrained. During transition, continuity and coupling strengthen, and small differences in forcing can produce large changes in behavior. Geometry enters not as description, but as constraint (Mandelbrot, 1983; Malamud et al., 1998). Once a coherent front forms, the shape of that boundary governs how the fire can grow. The system is no longer defined by local rules alone, but by the global structure those rules have produced. The central hypothesis of this project is that these regime transitions are not only real, but observable, measurable, and modelable.

Core products are empirical and computational: a curated event-level dataset with geometry and coupling metrics; open-source scaling and model-comparison diagnostics; a transition-aware benchmark suite spanning observations and simulations; and validated uncertainty reporting templates for decision-relevant interpretation. The intended outcome is not a single "winner" model, but a mechanism-discrimination framework that narrows plausible explanations, clarifies model limits, and provides a transparent path from physical hypothesis to robust comparative evidence (Sullivan, 2009).

The implication is direct: if wildfire growth is governed by changing regimes, then models must be evaluated not only by how well they fit outcomes, but by whether they represent the processes that control those regimes.

The goal is not to produce a single best model, but to determine which mechanisms are necessary under which conditions. The primary products are therefore both empirical and inferential: a curated event-level dataset with geometry and coupling metrics, a regime-classification system with uncertainty, and a benchmark suite that allows competing models to be evaluated against shared observational diagnostics. By linking physical hypothesis, observable structure, and model performance within a single workflow, the project provides a direct path from data to mechanism discrimination. In doing so, it aims to shift wildfire modeling from calibration against outcomes toward explanation of process.

2) Intellectual Merit

Target length: 225 words. The intellectual merit lies in unifying wildfire geometry, process dynamics, and uncertainty quantification within a falsifiable inferential framework. This requires three linked advances that build on one another. We first formalize measurable regime-transition diagnostics, including continuity, geometry, and transport-demand proxies, so we can test when local spread assumptions cease to be sufficient. We then develop a hybrid modeling architecture that combines local process realism with reduced geometric representations in regime-dependent ways. This progression culminates in a strict verification and validation pipeline that evaluates not only predictive error, but structural fidelity, identifiability, and calibration under changing forcing.

Methodologically, the proposal integrates event-based geospatial analytics, stochastic and mechanistic model experiments, and multi-objective comparison metrics including distributional and time-series diagnostics. It extends beyond pointwise forecasting by asking whether models reproduce observed cross-scale structure, such as perimeter-area scaling behavior, interface roughening, and regime-specific dynamics (Malamud et al., 1998; Mandelbrot, 1983). By explicitly defining failure criteria and reporting negative outcomes transparently, the project advances wildfire science as a mechanism-elimination program rather than a tuning exercise.

The deeper intellectual contribution is a shift in how wildfire models are judged (Sullivan, 2009). Instead of rewarding aggregate fit alone, we test whether models explain the observed organization of fire growth across scales and conditions. This reframing supports theory building, not just prediction contests.

The project also advances statistical practice in coupled Earth-system inference by integrating model comparison, calibration assessment, and structural diagnostics into a single hypothesis-testing loop. Rather than treating these as separate post hoc analyses, we use them as coordinated evidence streams that can jointly support or reject mechanism claims. This improves inferential rigor and makes it possible to separate apparent skill due to flexible fitting from robust explanatory power tied to observable fire-growth structure.

3) Broader Impacts

Target length: 225 words. Broader impacts center on translation, workforce development, and open scientific infrastructure. The project will release reproducible tools, benchmark datasets, and method documentation so that external groups can evaluate model behavior with a shared analytical language. This reduces duplication, improves comparability across studies, and lowers entry barriers for institutions that lack large internal modeling teams.

Training is embedded in project execution through structured roles for postdoctoral researchers and interdisciplinary collaboration across geospatial data science, complex systems modeling, and uncertainty analysis. Trainees will build publishable systems, lead conference outputs, and gain experience in transparent workflow design, strengthening the wildfire modeling workforce.

The project also includes public-facing translation pathways described in repository planning documents: interactive exploratory tools, educational materials, and narrative media partnerships designed to communicate mechanism-based wildfire understanding to broader audiences. The broader-impact goal is to move beyond one-way dissemination toward reusable, feedback-informed infrastructure that supports scientists, agency research groups, and long-term downstream decision systems.

We will evaluate impact through adoption and use, not only dissemination counts: external reuse of code and datasets, workshop participation outcomes, educational integration, and evidence that uncertainty-aware outputs change how wildfire modeling results are interpreted.

Broader-impact implementation is designed to be cumulative across project years. Early releases focus on transparent data and metric standards, mid-project releases emphasize training and interactive interpretation, and final releases package benchmark and communication assets for durable external use. This staged approach supports practical uptake while preserving scientific quality control.

Project Description (Target: 9,000 words total)

1. Introduction, Motivation, and Opportunity

Target length: 800 words. Wildfire science now operates in a paradox (Bowman et al., 2009). On one hand, data richness has expanded dramatically through satellite burned-area records, event reconstruction systems, and environmental reanalysis products. On the other hand, explanatory and predictive confidence remains fragile in the exact conditions where stakes are highest: rapid growth windows, compound forcing, and large coherent fronts (Abatzoglou and Williams, 2016; Balch et al., 2017). Existing model classes are often evaluated against local rates or final extent, yet those endpoints can conceal whether the model captured the process transitions that generated the event trajectory.

The difficulty is not a lack of data or computational power. It is a mismatch between how wildfire is represented and how it behaves. Many models are evaluated against local spread rates or final burned extent, yet these summaries can conceal whether the model captured the transitions that

produced the event (Rothermel, 1972; Finney, 1998). A fire that accelerates, reorganizes, and forms a coherent front is not simply a faster version of the same process. It is operating under different effective controls. If those controls are not represented explicitly, improved calibration cannot resolve the discrepancy.

This is not a marginal failure. It occurs in the conditions where prediction is most needed and most difficult. When fires reorganize, small differences in conditions can produce large differences in outcome. Models that assume a single governing process are least reliable precisely when the system changes most rapidly.

If wildfire growth is governed by changing regimes, then treating it as a single process is not a simplification—it is a misrepresentation (Sullivan, 2009; Finney et al., 2013). The problem is not that existing models are too simple or too complex, but that they are structured around the wrong assumption.

This gap creates a scientific opportunity. This shift is not only conceptual; it is measurable. Repository strategy documents emphasize a shift from "single-law" framing to a regime-based framing in which wildfire can transition between states with different effective controls. In pre-emergent phases, spread may be patchy, ignition-limited, and strongly local. During transition, continuity and acceleration can change rapidly as coupling strengthens. In post-emergent phases, interface organization and geometry may become dominant constraints. A valid model class should therefore be judged by regime-specific adequacy, not by aggregate fit alone.

This shift is not only a change in rate; it is a change in what controls the system. What matters early is not what matters later. Treating these as the same process obscures the mechanism that drives the transition.

The FIRE-MODEL track calls for innovative methods and algorithms across disparate spatiotemporal scales, measurable pathways for model improvement, and holistic verification and validation with uncertainty quantification. The consequence is a change in what must be modeled. Our design maps directly to these priorities. We build measurable transition diagnostics first, then test mechanism classes against those diagnostics, then package the workflow as reusable benchmark infrastructure. This sequence strengthens evidentiary clarity: what is observed, what is inferred, what each model must reproduce, and what constitutes rejection. This creates a natural question: which mechanisms can be simplified, and which must remain explicit, when the system changes most rapidly?

The project is also timely because foundational infrastructure already exists in this repository's scientific ecosystem: event reconstruction through `FIRED`, cube-based computational workflows through `CubeDynamics`, and an articulated methods ladder spanning measurement, model-data comparison, generative mechanism testing, and synthetic validation. The proposal does not begin from a blank slate; it leverages mature conceptual and computational scaffolding to pursue a decisive, falsifiable research program.

This creates a second constraint: model outputs are increasingly used in planning and response contexts, yet uncertainty communication remains underdeveloped. Overconfident interpretation is a practical risk under the conditions that matter most. By distinguishing reducible from irreducible uncertainty and linking reliability to regime context, we improve not only scientific understanding but the integrity of downstream interpretation.

This leads to a more specific problem: wildfire model development often proceeds with heterogeneous metrics, non-comparable preprocessing assumptions, and limited transparency around failure cases. The benchmark architecture proposed here is designed to make results portable and comparable across model families. This improves reproducibility and accelerates community learning.

Finally, this effort is positioned as a resubmission-strengthening program rather than a conceptual restart. The repository includes explicit reviewer-response framing: move from broad assertion toward testable discriminators, define decisive benchmarks, and make success/failure criteria operational. This proposal is the implementation of that strategy.

A final motivation is interoperability across wildfire research communities. Different groups often maintain different preprocessing assumptions, variable names, and validation conventions, which makes synthesis difficult even when findings are individually strong. By building a common event-centered representation and shared benchmark language, this project helps align model evaluation practice across institutions.

The opportunity, therefore, is not merely to produce one improved model, but to establish a durable scientific protocol: observable-first hypothesis testing, model role clarity by regime, transparent uncertainty decomposition, and reproducible comparison artifacts that can be carried forward as new datasets and models emerge.

Wildfire does not simply spread; it takes form, and that form reflects the processes that produced it (Mandelbrot, 1983; Malamud et al., 1998).

The project is explicitly theory-first generative modeling rather than simulation-first tuning. We begin from minimal governing principles that define wildfire growth as a process unfolding on a structured substrate, then test whether those principles are sufficient to reproduce observed fire forms. In this framing, fire form is treated as a measurable record of process: geometry, continuity, and scaling signatures are empirical constraints on mechanism validity, not decorative outputs (Kolmogorov, 1941; West et al., 1997).

This opportunity is amplified by rapid maturation in adjacent fields, including change-point detection, uncertainty-aware machine learning, and scalable geospatial data processing. These advances make it feasible to evaluate large wildfire event libraries with methods that were previously too computationally expensive or too fragmented to apply consistently. By integrating these capabilities into one reproducible system, the project can test regime hypotheses at the scale required for decisive evidence while maintaining transparent links between assumptions, data transformations, and conclusions.

2. Central Hypothesis and Specific Aims

Target length: 900 words. Central hypothesis. Large-fire growth is governed by transitions between regimes in which different mechanisms dominate. In early stages, growth is controlled by local spread and discontinuities in fuel and ignition. In later stages, once continuity is established, the advancing interface behaves as an organized structure whose geometry and coupling to the environment constrain further growth. The transition between these regimes can be detected through measurable changes in continuity, geometry, and transport-demand balance, and models that represent these transitions explicitly will outperform those that assume a single governing

process (Rothermel, 1972; Sullivan, 2009).

Aim 1: Detect regime transitions and their controls. We will construct and evaluate observable indicators from event-level perimeter dynamics, geometry metrics, and atmosphere-fuel proxy families (Malamud et al., 1998; Higuera et al., 2015). A core test follows the ventilation-capacity versus combustion-demand framing from the resubmission strategy documents: whether regime transitions cluster near thresholds in proxy ratios such as V/D. Candidate indicators include continuity measures, change points in scaling exponents, roughness evolution, and temporal alignment between transition markers and transport-demand crossings.

Aim 2: Explain transitions through model contrasts. We will implement and compare model families with distinct assumptions: local spread and wave-like formulations, reduced geometric and manifold formulations for coherent fronts, and hybrid transition-aware mixtures (Rothermel, 1972; Finney et al., 2013). This aim evaluates which mechanisms are necessary in each regime and which simplifications are defensible without losing structural fidelity. The focus is not only forecast accuracy, but mechanistic sufficiency under explicit comparison protocols.

Aim 3: Apply results as a reusable benchmark and inference system. We will release an uncertainty-aware benchmarking pipeline that allows other researchers to score model outputs against shared observational diagnostics. Deliverables include versioned datasets, metric libraries, evaluation templates, and reproducible workflows suitable for extension across ecosystems and future model classes.

Falsifiable predictions. (1) Observable transition signals will show reproducible change points across multiple datasets and regions. (2) Transition-aware model architectures will outperform single-regime baselines in out-of-sample transition-heavy events. (3) Calibration-aware uncertainty reporting will improve reliability diagnostics and expose model-form limits that are obscured by aggregate error summaries.

Success and failure logic. Each aim includes explicit decision thresholds. If transition indicators lack cross-dataset robustness, inferential claims will be narrowed to validated domains. If hybrid models fail to improve calibrated performance or structural fidelity in targeted regimes, we will reject broad transition-aware superiority claims. This explicit failure logic is central to proposal credibility and aligns with the repository's commitment to transparent reporting of negative or inconclusive results.

Hypothesis testing structure. At the core is "Benchmark 0" from the strategy documents: compare single-law and multi-regime explanations under identical data and validation conditions. We will use AIC/BIC and related information criteria, cross-validation, explicit change-point comparison, and transition-alignment analysis testing whether inferred regime boundaries co-occur with independent transport-demand transitions such as crossings in V/D proxy families. This creates a decisive test rather than rhetorical contrast.

Regime classifier as product. Beyond narrative claims, we will deliver a versioned regime classifier that labels fire states as pre-emergent, transition, or post-emergent with uncertainty. This classifier is designed to be usable both for observed events and for simulated outputs, enabling direct model-observation comparability.

Alternative outcomes. The project is designed to remain informative under multiple outcomes. If evidence supports regime transitions but weakly supports specific proxy families, we refine tran-

sition detection while treating mechanism proxies as provisional. If no robust transitions are found, we report conditions under which single-regime assumptions remain adequate. In all cases, the benchmark outputs and reproducible pipelines remain valuable community assets.

Experiment matrix and crosswalk. Each aim maps to a concrete experiment matrix. For Aim 1, we test indicator robustness across event classes, ecosystems, and sensing conditions. For Aim 2, we evaluate model classes under matched forcing and boundary definitions to isolate mechanism differences. For Aim 3, we package these tests into reusable benchmark workflows with explicit metadata and reporting templates. This crosswalk ensures that every narrative claim has a corresponding measurable test and artifact.

Decision-gate criteria. Advancement from one aim phase to the next depends on pre-specified criteria: stability thresholds for transition indicators, minimum calibration standards in out-of-sample evaluation, and reproducibility checks for pipeline outputs. Criteria are intentionally conservative to prevent over-interpretation of early signals.

A major synthesis output from these aims is a phase diagram of wildfire behavior in reduced regime space, centered on transport-demand ratio (for example V/D) and connectivity, with extension axes for forcing anisotropy, dryness or effective resistance, coupling strength, and heterogeneity scale ratio where data support them (Malamud et al., 1998; West et al., 1997). The phase diagram will classify fragmented, transitional, and coherent-continuous regimes with uncertainty bounds and will be released as both a figure-level and concept-level product.

The transport-demand ratio (V/D) expresses a simple physical idea: whether the atmosphere can remove heat and combustion products as quickly as the fire generates them (Kolmogorov, 1941; West et al., 1997). When that balance shifts, the fire's behavior must change.

Each aim also includes explicit replication tests. For transition detection, we require indicator stability under alternate preprocessing choices and across independent event subsets. For mechanism contrast, we require that leading model classes retain relative ranking under alternate calibration windows and random seed perturbations. For benchmark packaging, we require successful end-to-end reruns by project members not involved in initial development. These replication checks are included to reduce hidden analyst dependence and increase confidence that results are transferable beyond the originating team.

To strengthen causal interpretation, we will include controlled synthetic interventions where one mechanism axis is perturbed while others are held fixed. Examples include altering effective connectivity while preserving mean forcing, or rotating anisotropy while preserving magnitude. If predicted regime transitions shift in expected directions, confidence in mechanism attribution increases; if not, proposed controls are downgraded. This intervention logic complements observational inference and helps prevent over-interpretation of coincident correlations.

3. Data, Measurements, and Preprocessing

Target length: 1,100 words. The empirical backbone of the project is event-based reconstruction and multivariate coupling. `FIRED` provides coherent wildfire events from satellite burned-area pixels, including temporal progression and event-level perimeters. `CubeDynamics` provides the computational grammar for organizing and transforming spatiotemporal arrays, enabling reproducible

integration of perimeter trajectories with meteorology, fuels, and landscape covariates.

Our measurement system follows the repository's methods hierarchy. We begin by deriving canonical trajectory variables: $P(t)$, $A(t)$, dA/dt, and associated geometry descriptors. We then estimate scale-conditioned metrics such as perimeter-area relationships, roughness proxies, and anisotropy indicators (Malamud et al., 1998; Mandelbrot, 1983). What follows from this is interface-conditioned environmental coupling (e.g., wind, stability, moisture proxies), which lets us test coupled hypotheses instead of purely geometric correlations.

These measurements are not descriptive in the usual sense; they are the observables through which regime transitions can be detected and tested. Each quantity is chosen because it constrains whether growth is locally controlled, transition-driven, or interface-dominated.

To unify observations and simulations, we represent growth using an ignition-time field $T(x, y)$, or equivalent arrival-time representation, defined on harmonized spatial grids. This common representation jointly encodes front position, spread progression, and geometry through time, enabling direct computation of diagnostics and direct comparison between observed and simulated fires without ad hoc output translation.

Data quality and uncertainty are treated as first-class modeling inputs. Every transformation step will preserve provenance metadata, quality flags, and versioned configuration. Missingness is addressed through sensitivity designs comparing complete-case analysis, principled imputation, and uncertainty-weighted estimators. Sensor and resolution dependence are quantified through perturbation analyses, allowing us to separate robust signatures from measurement artifacts.

The preprocessing architecture is intentionally modular. Components for perimeter extraction, geometry estimation, environmental coupling, and metric calculation are containerized as reproducible pipeline stages with documented interfaces. This allows partial progress even when one data stream is delayed and supports external reuse by collaborators who may only require subset functionality.

Deliverables from this component include Dataset v1 and v2 artifacts consistent with repository planning language: event-level time series, derived observables, uncertainty envelopes, and reproducible pipeline code with benchmark-ready outputs.

We will harmonize temporal alignment across data sources using explicit event clocks and resampling windows. This is necessary because sensor products, meteorological reanalysis, and fuel datasets operate at different temporal granularity and reporting conventions. Harmonization procedures will be documented and sensitivity-tested so that modeling outcomes are not artifacts of arbitrary alignment choices.

Measurement definitions will be explicitly versioned. For each key metric, we will track formula choices, smoothing assumptions, spatial discretization parameters, and treatment of disconnected geometry. This addresses a recurring comparability problem in wildfire analysis: two studies may report the same named metric while using materially different definitions.

Environmental coupling will be handled through interface-conditioned sampling rather than only event-level averages. This preserves directional and local context and supports tests of whether particular boundary segments, rather than whole-fire means, are associated with transition behavior.

Quality control includes anomaly detection on trajectory continuity, geometric plausibility checks, and cross-source consistency tests. Events that fail minimum quality criteria will be flagged with transparent exclusion logic rather than silently dropped. We will maintain an audit trail of exclusions so downstream inference remains interpretable.

The final data products are designed for dual use: immediate project inference and external benchmark adoption. To support this, we will provide clear schemas, example notebooks, and machine-readable metadata that describe variable definitions, units, uncertainty fields, and processing lineage.

Data stewardship considerations are integrated from the start. For each source, we will track access constraints, usage permissions, and citation requirements, and we will maintain clear separation between openly shareable derivatives and restricted components. This allows transparent compliance while maximizing reproducibility where licensing permits.

Computational reproducibility is treated as a measurable output. Pipeline runs will capture configuration snapshots, dependency manifests, random seeds where applicable, and runtime provenance metadata. This makes it possible to reproduce benchmark outcomes and to audit differences when results diverge between environments.

Event selection rules will be pre-registered at the analysis-plan level. Inclusion criteria will specify minimum temporal duration, minimum reconstructed growth signal, and data quality thresholds for geometric continuity. Exclusion criteria will include unresolved sensor ambiguity, severe temporal gaps that prevent transition inference, and boundary artifacts that cannot be corrected under documented procedures. Pre-registration of these rules limits retrospective selection bias and clarifies the inference population represented by benchmark outputs.

We will also build a proxy-construction framework for transport-demand and coupling variables that supports side-by-side comparison of alternative formulations. For each proxy family, we will track physical interpretation, data dependency, and sensitivity to resolution. This prevents hidden mixing of conceptually distinct quantities under one label and enables transparent tests of whether transition signals are robust to proxy definition.

Cross-scale consistency checks will connect pixel-level observations to event-level trajectories. We will evaluate whether event summaries preserve key local signatures and whether local anomaly patterns aggregate into meaningful transition indicators. Where inconsistencies appear, we will diagnose whether they arise from discretization, missingness, or true multi-scale process differences. This supports defensible interpretation when moving between local and global views of fire behavior.

Uncertainty propagation will be implemented directly in measurement outputs, not appended only at the reporting stage. Derived variables will include uncertainty fields generated from source-data confidence, interpolation assumptions, and metric-estimation sensitivity. These fields will be consumed by downstream model comparison so that uncertain observations do not carry the same weight as high-confidence observations in mechanism evaluation.

To facilitate external adoption, data products will be released in layered formats: raw harmonized inputs where permissions allow, canonical processed tables for immediate analysis, and compact benchmark bundles for computationally constrained users. Documentation will include clear examples for each layer so users can choose the entry point that matches their technical and resource constraints.

We will include domain-specific data cards for each major benchmark release. These cards will summarize geographic coverage, ecosystem representation, known sensing constraints, and expected applicability limits. Data cards provide fast interpretive context for users and reduce the risk that benchmark outputs are applied outside the domains where measurement confidence and inferential support are adequate.

Release cadence for data products will follow a clear stability policy. Development snapshots support rapid iteration, while milestone-tagged releases lock definitions for comparative evaluation and manuscript use. This policy reduces ambiguity about which data version underlies any reported result and helps external users reproduce claims without reverse-engineering intermediate pipeline states.

4. Modeling Framework and Algorithmic Innovation

Target length: 1,300 words. The project evaluates wildfire dynamics through complementary model roles rather than a single monolithic simulator (Finney, 1998; Finney et al., 2013). The first role is local spread representation: process-informed models that capture ignition propagation, local coupling, and conditional heterogeneity effects. The second role is coherent-front representation: reduced geometric formulations that focus on boundary evolution once front continuity emerges. The third role is transition inference: statistical mechanisms that detect regime state and mediate handoff or weighted mixing between model roles.

This is a shift in what counts as explanation. A model that reproduces total burned area may still fail to explain the fire if it does not reproduce the structure of the advancing boundary (Sullivan, 2009). Agreement in outcome is not sufficient; agreement in form is required. Because form is the trace left by process, it provides a more stringent test of mechanism than aggregate statistics alone.

If this perspective is correct, then the goal is not to build a more detailed version of the same model, but to represent the change in what must be modeled. The question is not how to simulate fire more precisely, but how to identify which processes matter at each stage of growth.

This architecture is motivated by repository narrative synthesis and resubmission strategy. In pre-emergent and transition regimes, local spread assumptions are often indispensable; in post-emergent regimes, boundary geometry may explain a larger fraction of growth variability (Rothermel, 1972; Finney et al., 2013). The project therefore asks not "which model wins globally" but "which model class is sufficient under which regime conditions."

This shift is not only conceptual; it is measurable. Wildfire does not simply spread; it takes form, and that form reflects the processes that produced it (Mandelbrot, 1983; Malamud et al., 1998).

The modeling philosophy is explicitly theory-first and generative. The consequence is a change in what counts as adequate model fit. We begin from minimal governing principles and test whether they are sufficient to reproduce observed fire form. In this framing, form is not a byproduct of simulation; it is evidence. The geometry and topology of the advancing interface encode the history of the process that produced them. If a model reproduces burned area but fails to reproduce interface structure, scaling behavior, or continuity transitions, it has not captured the underlying mechanism. This shifts evaluation from agreement in outcomes to agreement in structure.

As a scientific instrument for mechanism discrimination, we use a staged model hierarchy (Rothermel, 1972; Scott and Burgan, 2005; Finney et al., 2013). M0 represents isotropic, fuel-controlled growth. M1 adds directional anisotropy from wind and slope. M2 adds coupling terms that organize and stabilize the advancing front. M3 adds extended dynamics where justified by evidence, including nonlocal effects and richer transport-demand coupling. This hierarchy is not a convenience layer; it is the primary instrument for identifying which added ingredients are necessary in which regimes.

Across this hierarchy we seek reduction into a compact regime space rather than uncontrolled parameter growth. Candidate reduced controls include connectivity, dryness or effective resistance, forcing anisotropy, coupling strength, and heterogeneity scale ratio (Kolmogorov, 1941; West et al., 1997). Final dimensionless forms will be data-constrained, but the conceptual objective is explicit: reduce apparent complexity into a small number of testable controls.

Algorithmic innovation appears in four areas.

(i) Transition-aware gating and mixtures. We compare deterministic thresholds, probabilistic gating, and continuous-mixture formulations based on transition indicators. These alternatives are evaluated on both predictive metrics and interpretability of inferred state changes.

(ii) Mechanism-elimination experiments. Guided by the repository's generative framework, we stage model experiments that test necessary ingredients: dimensional adequacy, interface roughening, connectivity constraints, and geometry-throughput coupling. Models are retained only when they satisfy both exponent-level and geometry-level criteria within uncertainty.

(iii) Surrogate and emulator acceleration. We develop validated surrogates for expensive model components to support broad parameter sweeps and uncertainty decomposition. Surrogate use is constrained by strict domain-of-validity checks against reference runs.

(iv) Cross-model comparability. All model outputs are converted into common observable space so that comparisons occur on aligned measurements rather than model-internal variables. This reduces ambiguity and supports fair ranking.

The expected outcome is a constrained mechanism map: where local physics is necessary, where reduced geometry is sufficient, and where hybrid formulations provide the strongest explanatory and predictive tradeoffs.

Core figure outputs are named in advance to improve reviewability and execution clarity: the decisive single-law versus multi-regime comparison figure (Benchmark 0); a regime-classification time series with uncertainty; a transition-alignment figure comparing observed transition timing with V/D crossings; a phase diagram in reduced regime space; and a cross-model comparison panel stratified by regimes including transition and extreme-fire windows.

Model development will proceed in explicit tiers. Tier 1 focuses on robust baseline implementations and reproducible parameterization. Tier 2 introduces regime-dependent augmentation and transition logic. Tier 3 performs stress testing under extreme scenarios and cross-domain transfer checks. This staged design reduces integration risk and keeps evaluation synchronized with model complexity.

We will benchmark against established and commonly referenced model classes represented in

repository documents, including operational and research simulators where compatible outputs are available, as well as statistical and null baselines (Finney, 1998; Finney, 2004; Sullivan, 2009). The goal is not to claim universal superiority, but to identify condition-specific advantages and failure modes under transparent protocols.

A central innovation is mechanism sufficiency mapping. Instead of ranking only by aggregate score, we ask which mechanism families remain consistent with all validated constraints in each regime class. This creates a sharper scientific product: elimination of implausible explanations and bounded support for survivors.

We also formalize geometry-to-throughput coupling tests that link interface properties to growth rates. Competing formulations, such as dA/dt scaling with perimeter alone versus perimeter interacting with environmental state, will be compared with cross-validated and information-criterion approaches. This links geometric claims to dynamical consequences.

Extreme-event behavior receives dedicated attention (Finney et al., 2013; Abatzoglou and Williams, 2016). Many models can perform adequately in moderate conditions yet fail in directional, high-growth windows. We therefore include targeted extreme-regime tests using anisotropy, rapid-acceleration, and forcing alignment diagnostics. Mechanisms that fail these tests are not promoted as general explanations.

Finally, implementation choices prioritize portability. Model wrappers, configuration schemas, and output adapters will be designed so additional model families can be integrated with limited custom engineering. This supports long-term benchmark evolution beyond the grant window.

To prevent methodological drift, every model branch will be evaluated under harmonized input definitions and output extraction rules. Where models require distinct internal parameterizations, comparability will be preserved through standardized external observables and common scoring protocols.

The project also includes ablation-style experiments that selectively remove candidate mechanism components, such as connectivity constraints or environmental coupling terms, to test whether specific ingredients are necessary for reproducing observed structure. These tests help move from correlative fit toward causal plausibility.

Parameter inference will combine interpretable calibration with robustness-oriented exploration. We will use regime-stratified calibration windows to avoid fitting one set of parameters across incompatible dynamical conditions, and we will evaluate whether parameter posteriors remain stable under alternate event subsets. Instability will be treated as evidence of model misspecification or hidden regime dependence rather than absorbed as routine variance.

We will compare deterministic and probabilistic model forms under consistent scoring rules. Deterministic forms will be assessed for structural adequacy and bias patterns, while probabilistic forms will be assessed for both central tendency and calibrated uncertainty. This side-by-side design ensures that apparent gains from stochastic flexibility are interpreted alongside explanatory clarity and identifiability.

Model integration infrastructure will include standardized adapters for input harmonization, run control, and output export into shared observable space. These adapters reduce one-off engineering overhead and make cross-model experiments tractable at project scale. They also make it easier to

include additional external models in later benchmark versions without rewriting the comparison pipeline.

Where computational cost is high, we will use adaptive experiment design to focus simulation effort on informative regions of parameter and forcing space. Rather than uniformly sampling all combinations, the design iteratively allocates new runs where model disagreement is highest or where transition boundaries are most uncertain. This increases inferential efficiency while preserving reproducibility through explicit decision logs.

A final algorithmic objective is principled simplification. If reduced formulations can match structural and uncertainty-aware diagnostics in defined regimes, they will be promoted for those regimes even when more complex alternatives are available. This prevents unnecessary model complexity and supports clearer mechanism interpretation.

Finally, we will maintain a living model-phylogeny record that documents how each implemented model class relates to prior formulations and to the M0–M3 hierarchy. This record supports reproducible comparison by making assumptions explicit and by clarifying where model families are genuinely distinct versus minor parameter variants.

Interpretability is treated as an algorithmic criterion, not only a communication preference. For each model family, we will document which parameters map to physical controls, which represent statistical flexibility, and which are auxiliary calibration terms. This mapping allows reviewers and users to understand why models succeed or fail in specific regimes and prevents conflating predictive convenience with mechanistic adequacy.

5. Verification, Validation, and Uncertainty Quantification

Target length: 1,200 words. Verification and validation (V&V) are designed as the scientific core of this proposal, not an afterthought. Repository planning documents emphasize that success should be defined by reliable mechanism discrimination and transparent uncertainty, including in negative-result scenarios. Our V&V framework therefore combines predictive testing, structural diagnostics, and uncertainty calibration under shared protocols.

In this project, validation is not a final check on model performance; it is the primary scientific test through which competing explanations are accepted or rejected.

Wildfire does not simply spread; it takes form, and that form reflects the processes that produced it (Mandelbrot, 1983; Malamud et al., 1998). What follows from this is a stricter validation requirement: models must reproduce regime-specific structure, not only final totals (Mandelbrot, 1983; Sullivan, 2009).

A model that succeeds under stable conditions but fails during transition is not incomplete; it is misleading (Finney et al., 2013; Sullivan, 2009). The value of this framework lies in identifying those failures explicitly.

Predictive performance. We evaluate event-level and time-local performance under cross-validation and held-out scenario tests, with special emphasis on transition-heavy windows. Metrics include trajectory error, distributional divergence, and time-warp-aware alignment where appro-

priate.

Structural fidelity. Because aggregate error can mask mechanism failure, we require models to reproduce regime-relevant structure: continuity changes, roughness evolution, perimeter-area behavior, and anisotropy signatures. This is where benchmark language from repository model pages is directly operationalized.

Uncertainty calibration. We decompose uncertainty into observational, parametric, and model-form components. Calibration diagnostics evaluate whether uncertainty intervals are informative and reliable, not merely wide. Reporting will distinguish uncertainty that may be reduced through better data/modeling from uncertainty likely intrinsic to regime variability.

Identifiability and robustness. Synthetic validation experiments (with known generative conditions) test whether the full pipeline can recover intended mechanisms and avoid false confidence. Global sensitivity analyses evaluate whether "successful" behavior requires fragile tuning or persists across meaningful parameter ranges.

Failure reporting. If mechanisms cannot be distinguished within uncertainty, or if findings do not generalize across datasets, those outcomes are reported as substantive results. This aligns with repository evaluation language and strengthens scientific integrity.

Verification tasks will include implementation consistency checks, unit and integration test coverage for key metric modules, and reproducibility checks across compute environments where feasible. Validation tasks then operate on scientific behavior, not merely code execution. This separation reduces silent technical bias in scientific conclusions.

For model-data comparison, we will use a portfolio of complementary metrics rather than a single score. Repository planning materials already reference measures such as RMSE, KS distance, DTW alignment, and Earth Mover's Distance. These metrics capture different error structures and reduce overfitting to one objective.

We will explicitly test sensitivity to data definition choices, including boundary extraction rules, temporal aggregation windows, and smoothing settings. If conclusions depend strongly on one preprocessing variant, claims will be downgraded until robustness is demonstrated.

Uncertainty communication outputs will include visual and tabular templates designed for both technical and semi-technical audiences. The purpose is not only statistical completeness but interpretation discipline: users should understand what is known, what is uncertain, and what decisions should remain conditional.

Generalization is evaluated across event classes, geographies, and forcing regimes where data coverage permits. Where generalization is weak, we will report applicability domains rather than broad claims.

Validation reporting will be organized in tiers for different audiences. Technical reports will include full metric distributions, sensitivity results, and calibration diagnostics. Synthesis reports will provide concise decision-oriented interpretations with clear caveats. This two-tier approach preserves rigor while improving usability.

We will also maintain a "known limitations" registry that is updated as evidence accumulates.

This registry will document unresolved ambiguities, conditions with weak support, and model classes that fail specific criteria. Maintaining this registry as a first-class product supports scientific transparency and helps external users avoid overgeneralization.

Proposal-level acceptance criteria are stated directly. Evidence should favor multi-regime over single-law explanations in a majority of tested fires within analyzed domains; inferred regime transitions should align with V/D transitions within a predefined temporal tolerance; reduced geometric models should reproduce post-emergent behavior within stated uncertainty; and the benchmark suite should differentiate models on transition and extreme-fire behavior, not only aggregate fit.

If these criteria are not met, the negative result remains a substantive contribution: it constrains plausible mechanism space, bounds proxy usefulness, and clarifies limits of current wildfire modeling frameworks.

Calibration evaluation will include reliability diagrams, coverage-versus-width analysis, and threshold-conditioned performance summaries so that uncertainty quality can be interpreted in context rather than as a single aggregate statistic. We will also report how calibration changes across regimes and event sizes, since a model that is well calibrated in moderate conditions may be overconfident in extreme windows.

For identifiability, we will apply profile-likelihood and posterior-correlation diagnostics to detect parameter combinations that are effectively non-identifiable under available data. When non-identifiability is detected, claims tied to those parameters will be narrowed, and model simplification or reparameterization will be considered before proceeding to broad interpretation.

Cross-dataset transportability tests will evaluate whether model-form conclusions hold when moving across regions and sensor contexts. These tests are important because wildfire processes and data quality can vary substantially by ecosystem and monitoring regime. Transport failures will be documented explicitly as boundary conditions on claim validity.

To guard against optimistic bias from iterative development, we will reserve strict holdout sets that are not used for routine tuning. Holdout analysis will be scheduled at predefined checkpoints and documented in versioned reports. This creates a clear distinction between development performance and final evidentiary performance.

All major validation outputs will be delivered in machine-readable and human-readable formats. Score tables, uncertainty diagnostics, and failure annotations will be exportable for external analysis, while narrative synthesis summaries will interpret implications for model development priorities and applicability limits.

We will also conduct temporal stress tests that isolate known difficult windows such as abrupt acceleration periods, directional shifts under changing wind alignment, and late-stage coherent-front propagation. Reporting these windows separately from full-event summaries helps determine whether model adequacy in aggregate metrics is masking systematic failure in scientifically critical phases.

In addition, the validation workflow will include structured error taxonomy reporting. Errors will be categorized by type (timing, geometry, magnitude, directional bias), by regime, and by event context. This taxonomy turns validation into an actionable learning system by pointing directly to mechanism classes or data assumptions that most likely explain recurring failure patterns.

Validation governance will include preregistered scorecard templates that define primary metrics, secondary diagnostics, and interpretation thresholds before full comparison results are inspected. This limits metric-shopping and keeps comparison conclusions tied to predeclared criteria. Where post hoc exploratory analyses are performed, they will be clearly labeled as exploratory rather than confirmatory.

We will also run blind re-analysis exercises in which a second team member reproduces selected evaluation results from released artifacts without access to intermediate working files. Differences between original and reproduced outcomes will be documented and resolved before public release. This practice strengthens confidence in both computational reproducibility and reporting accuracy.

Where model comparisons remain statistically ambiguous, we will report evidence tiers rather than binary winners. Tier 1 evidence indicates robust separation across primary metrics and sensitivity checks; Tier 2 indicates conditional separation under specific assumptions; Tier 3 indicates no reliable separation. This tiered reporting prevents forced conclusions and helps prioritize future data collection or model development where uncertainty is currently dominant.

6. Work Plan, Milestones, and Deliverables

Target length: 1,000 words. The work plan follows the Detect–Explain–Apply sequencing already established in repository research-program materials and is implemented as decision-gated annual phases.

Year 1 (Detect). The objective is to determine whether regime transitions can be detected robustly across datasets, and whether the proposed indicators produce consistent and interpretable signals. Milestones include Dataset v1 with core observables ($P(t)$, $A(t)$, dA/dt, geometry, anisotropy), an initial comparison engine, and a first model-selection report contrasting single-regime versus multi-regime framing.

Year 2 (Explain). The objective is to determine which mechanisms are necessary to reproduce observed transitions, using controlled model contrasts across the M0–M3 hierarchy. Milestones include Model Library v1, Validation Framework v1, and evidence-weighted regime-specific sufficiency maps.

Year 3 (Apply). The objective is to formalize these results into a reusable benchmark system and to test whether the framework generalizes across fire regimes and model classes. Milestones include final benchmark artifacts, reproducibility bundles, and a consolidated mechanism synthesis report.

Deliverables are tracked as two coordinated streams. Scientific deliverables include transition tests, mechanism discrimination outcomes, and regime-specific inference results. Infrastructure deliverables include versioned datasets, pipeline code, benchmark utilities, and documentation packages. This dual-stream design ensures durable value even under partial or mixed scientific outcomes.

Explicit artifacts include a fire growth trajectory dataset derived from satellite observations, versioned regime labels with uncertainty, and V/D and related transport-demand proxy datasets. Additional artifacts include CubeDynamics-integrated scaling diagnostics, a model experiment library covering M0–M3 and ablation variants, a benchmark suite for existing and future fire models, and a

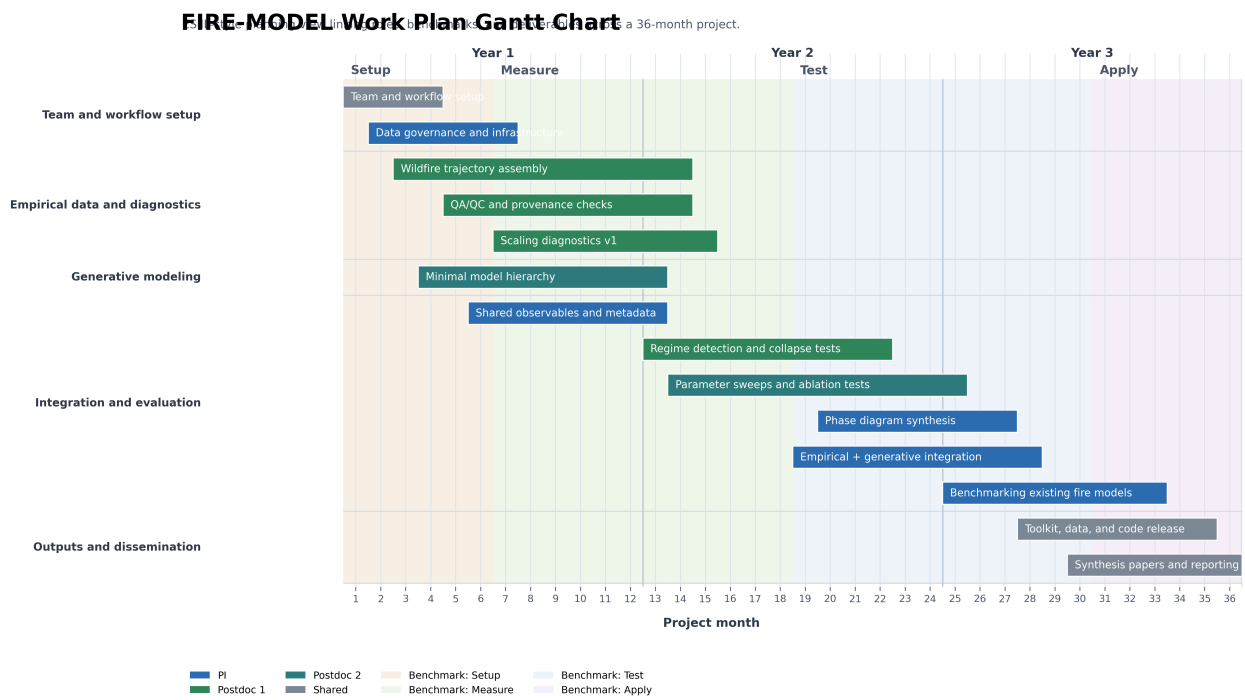

Figure 1: FIRE-MODEL three-year Gantt timeline generated from the repository planning workflow.

consolidated theoretical framework for wildfire growth geometry anchored in measurable evidence.

Management cadence mirrors repository staffing and evaluation plans: weekly discrepancy reviews across measurement-model-validation teams, quarterly milestone assessments, and explicit risk tracking with contingency triggers for data bottlenecks, computational delays, and underperforming model branches.

Within each year, we define quarter-scale decision points. Early quarters emphasize pipeline stability and measurement reliability. Mid-year quarters emphasize comparative analysis and model discrimination. End-of-year quarters emphasize synthesis, release readiness, and conference manuscript alignment. This quarterly structure prevents milestone drift and supports corrective action before annual reporting windows.

Benchmark 0 is an explicit work-plan gate rather than background context. It is executed as a formal deliverable with AIC/BIC comparison, cross-validation, change-point analysis, and alignment tests against transport-demand transitions. Broad claims about regime structure are contingent on passing this gate.

Conference and manuscript outputs are tied to milestone maturity, not arbitrary dates. For example, early measurement papers depend on Dataset v1 stability; model-comparison papers depend on completion of cross-model evaluation and uncertainty diagnostics. This improves coherence between narrative outputs and demonstrated evidence.

Risk governance includes explicit fallback plans. If hiring timelines shift, core pipeline and benchmark foundations will be advanced by PI and collaborator effort using modular task decomposition.

If data coverage is uneven, we prioritize robust domain-specific claims over weak generalization.

We will maintain a live milestone dashboard in repository documentation to track status, blockers, and evidence artifacts for each deliverable. This supports internal accountability and efficient resubmission updates.

Deliverable packaging includes explicit handoff standards: each major artifact must include a plain-language summary, technical specification, reproducibility instructions, and validation status. This reduces integration friction across workstreams and improves external reuse.

At project close, we will deliver a consolidated evidence book that links hypotheses, diagnostics, model tests, uncertainty findings, and milestone artifacts. This product is intended to make reviewer and stakeholder evaluation efficient and transparent.

Operationally, each year is partitioned into initiation, integration, and synthesis phases. Initiation phases prioritize technical setup, data integrity checks, and baseline reproducibility. Integration phases prioritize cross-team experiment execution and discrepancy resolution. Synthesis phases prioritize interpretation, documentation, and release packaging. This repeated phase pattern makes annual progress legible and supports early detection of schedule drift.

Year 1 quarter-level goals include event catalog stabilization, metric-definition freeze for core diagnostics, and completion of Benchmark 0 with at least one cross-region replication. Year 2 quarter-level goals include completion of ablation experiments, calibration audits across regimes, and publication-ready comparison figures. Year 3 quarter-level goals include external user testing of benchmark workflows, final uncertainty-reporting templates, and public release readiness checks.

We will maintain dependency maps for critical milestones so that blocking tasks are visible before deadlines. These maps will link data readiness, model readiness, and validation readiness for each major deliverable. If bottlenecks appear, mitigation actions such as scope narrowing, phased release, or temporary staffing reallocation will be triggered under predefined rules.

A dissemination calendar is integrated into the work plan. Internal milestones feed manuscript drafting windows, conference abstract deadlines, and workshop preparation. This alignment ensures that outward-facing outputs are generated from validated artifacts rather than speculative intermediate results.

Final-year application work includes packaging, onboarding, and transfer tasks. Packaging covers benchmark release engineering and archival documentation. Onboarding includes tutorials and support sessions for early external adopters. Transfer includes explicit handoff materials that allow the benchmark system to persist and evolve beyond the project end date.

Annual retrospective reviews will document whether planned milestones were met, which assumptions required revision, and which methodological choices produced the largest gains in robustness. These retrospectives will be used to tune subsequent-year priorities and to ensure that schedule pressure does not override evidence-based adjustment.

Each major milestone will have explicit entry and exit criteria. Entry criteria ensure prerequisite data and tooling are stable enough for meaningful comparison. Exit criteria require complete documentation, reproducibility checks, and uncertainty summaries in addition to core scientific results. This gate design prevents premature progression and improves downstream synthesis quality.

Schedule reserves are built into each year for integration debt and unforeseen technical tasks. These reserves are not idle buffers; they are designated time blocks used for reconciling cross-team interfaces, improving documentation quality, and resolving reproducibility issues before public-facing outputs are finalized.

7. Team Integration, Convergence, and Management

Target length: 700 words. The team structure follows the repository staffing model: two complementary postdoctoral tracks and PI-level integration. One track focuses on computational geospatial measurement and event-level data quality. The second focuses on complex systems modeling and mechanism experiments. The PI and senior collaborators coordinate convergence logic, milestone governance, and synthesis.

Convergence is defined operationally, not rhetorically. No claim advances without aligned evidence across three artifacts: measured observables, model outputs in common metric space, and uncertainty diagnostics. This artifact-centered review process reduces disciplinary drift, enforces comparability, and creates auditable reasoning chains from data to conclusion.

Role-specific outputs map to project milestones. Measurement leadership delivers dataset versions, geometry diagnostics, and uncertainty envelopes. Modeling leadership delivers mechanism libraries, comparative experiments, and transition-aware formulations. Integration leadership delivers cross-model evaluation, risk resolution, and proposal-level synthesis products. Annual outputs include conference dissemination and first-author manuscript progression, as specified in planning documents.

Risk mitigation is built into modular dependencies: measurement, modeling, and validation can proceed in parallel with iterative synchronization. If one stream slows, other streams generate actionable diagnostics that guide adaptation rather than causing full stoppage.

Team operation includes weekly technical integration meetings, monthly synthesis meetings, and quarterly milestone reviews. Weekly meetings resolve discrepancies between measurement and modeling outputs. Monthly meetings evaluate scientific implications and adjust experiment priorities. Quarterly reviews confirm milestone completion against predefined acceptance criteria.

Mentoring and professional development are explicit project responsibilities. Postdoctoral researchers will lead major deliverables, manage reproducible workflows, and present at conferences as first authors where appropriate. This approach builds workforce capacity while strengthening project execution.

External advisory input will be used selectively for methods and interpretation checkpoints, especially around uncertainty communication and benchmark usability. Feedback will be documented and translated into explicit action items.

Coordination protocols include explicit interfaces between personnel roles. Measurement teams provide versioned data products on scheduled release cycles; modeling teams provide benchmark-ready outputs with standardized metadata; validation teams provide comparative scorecards and uncertainty summaries. These interfaces reduce coordination ambiguity and make dependencies testable.

The management structure also emphasizes continuity planning. Key workflows will be documented at a level that allows role transitions without loss of institutional memory, supporting resilience under staffing changes and improving long-term project sustainability.

Convergence management also includes explicit conflict-resolution protocols for scientific disagreements. When model and measurement teams produce divergent interpretations, the default response is targeted diagnostic testing under pre-agreed criteria rather than informal consensus. This keeps project decisions evidence-centered and prevents timeline slippage caused by unresolved conceptual debates.

Training objectives are mapped to concrete competencies, including reproducible geospatial workflow construction, uncertainty-aware modeling, model-comparison interpretation, and scientific communication for mixed audiences. Progress on these competencies will be reviewed semi-annually and tied to role evolution, manuscript leadership, and conference representation.

The project will maintain a lightweight governance record that logs major methodological decisions, rationale, and downstream implications. This record improves transparency, supports reproducibility, and helps new team members rapidly understand why specific modeling choices were adopted.

To support inclusive collaboration, team documentation and meetings will use structured agendas, explicit action ownership, and written follow-up notes. These practices reduce coordination overhead, make expectations clearer, and improve continuity when participants rotate across project tasks.

Team integration success will be assessed with lightweight process metrics, including turnaround time for cross-team discrepancy resolution, proportion of deliverables meeting reproducibility standards on first review, and continuity of milestone progress during personnel transitions. Monitoring these indicators provides early warning for management bottlenecks before they affect scientific outputs.

Cross-institution coordination will be supported through shared templates for data handoff, model run manifests, and validation summary reports. Standardized templates reduce avoidable interpretation errors and make it easier to compare contributions from different collaborators under consistent expectations.

The team will also conduct periodic role-rotation exercises for selected workflow components so that critical tasks are not concentrated in single individuals. Role rotation improves resilience, supports mentoring, and reduces operational risk if staffing changes occur during key project phases.

8. Broader Impacts Plan

Target length: 700 words. Broader impacts are structured as an integrated system rather than isolated outreach activities. Repository planning materials identify three coordinated components: an interactive wildfire exploration platform, narrative media and educational products, and an open science training ecosystem.

The practical consequence is not only improved models, but improved understanding of when those models can be trusted. A researcher comparing alternative approaches should be able to see not

only which model fits better, but where and why each model fails as conditions change.

Interactive component. The Fire Lab concept provides a user-facing interface for exploring model behavior, uncertainty, and scenario structure. Its purpose is not operational command support, but interpretable scientific engagement for students, researchers, and agency technical audiences.

Narrative and education component. Partnership with Impact Media Lab supports production of documentary-style and instructional materials that translate technical wildfire modeling ideas into accessible narratives while preserving scientific fidelity. Companion lesson materials connect conceptual framing, visual outputs, and mechanism-testing logic.

Open science component. Code, workflows, and benchmark assets are released with documentation and training modules. Workshops combine conceptual orientation, hands-on pipeline use, and feedback capture, creating bidirectional learning between developers and users.

Evaluation will use concrete indicators: workshop participation and outcomes, tool adoption, repository usage metrics, external method reuse, and evidence of educational uptake. The broader-impact test is whether transparent, reusable mechanism-based tools improve both scientific literacy and practical interpretation of wildfire modeling outputs.

Broader impacts are designed for continuity across audience types. Scientific users access reproducible methods and benchmark resources. Technical agency users access interpretable diagnostics and uncertainty templates. Educational users access structured narrative and instructional material. These layers are linked so that one set of outputs supports multiple pathways.

User pathways are defined explicitly. Direct users are fire model developers, remote sensing scientists, fire behavior researchers, and Earth system scientists. Secondary users are agency research groups and strategic planning or risk-assessment teams. The operational pathway is intentionally described as indirect: scientific insight $\rightarrow$ improved models $\rightarrow$ better forecasting tools $\rightarrow$ downstream operational benefit.

Workshops serve both dissemination and co-design functions. Participants will test tools, report friction points, and suggest scenario classes that matter for practice. This feedback will inform subsequent platform and documentation updates, creating an iterative improvement loop.

We will prioritize accessibility in communication artifacts through clear visual conventions, plain-language summaries, and modular learning materials that can be adapted for different educational contexts. These choices are intended to broaden participation in wildfire modeling literacy.

Long-term impact is defined as durable infrastructure and reusable practice: tools that continue to be used, workflows that are adopted, and conceptual frameworks that shape subsequent wildfire model development.

To support durability, we will align dissemination formats with likely user pathways: repository releases for technical users, structured curricula for educators, and concise narrative products for broader public audiences. These formats draw from the same underlying evidence base, reducing contradiction between technical and public-facing outputs.

We will assess sustainability through post-release indicators, including updates by external contrib-

utors, replication of workshop exercises, and reuse of benchmark templates in independent studies. These indicators provide a practical test of whether broader impacts persist beyond initial launch.

Broader-impact activities will be scheduled to align with technical maturity. Initial outreach emphasizes conceptual literacy and transparent uncertainty communication, while later activities emphasize hands-on benchmark use and adaptation. This sequencing avoids asking external users to engage immature tools and improves the quality of adoption feedback.

We will also produce modular educator-facing kits built from project figures, plain-language explanations, and guided activities. Kits will be designed for flexible use in undergraduate and graduate settings, with clear notes on prerequisites and suggested discussion prompts. This format supports repeat use and reduces preparation barriers for instructors.

Evaluation findings from workshops and external users will feed back into product design through a documented revision loop. Each cycle will capture observed usability issues, prioritized improvements, and release notes that explain what changed and why. This loop makes broader impacts measurable as iterative system improvement rather than one-time dissemination.

To sustain impact beyond the award period, we will prioritize open formats and clear governance for maintenance. Public-facing assets will include version histories, contribution guidance, and transparent update policies so that external partners can continue using and improving materials after the initial project window.

9. Results from Prior NSF Support and Positioning

Target length: 500 words. This section will be finalized with award-specific factual language at submission time. The draft framing is as follows: prior work established key enabling infrastructure, including event-level fire reconstruction workflows, scalable data-cube analytics, and methodological prototypes for geometry-aware evaluation. Those achievements revealed both opportunity and limitation: rich data and computational tools exist, but mechanism discrimination under regime change remains underdeveloped.

The present proposal is positioned as the next logical step: move from descriptive and component-level advances to a falsifiable, integrated inferential system that can test regime hypotheses, compare model classes under aligned diagnostics, and report uncertainty with decision-relevant clarity.

Positioning strength comes from the repository's existing integration: funder-aligned requirement tracking, detailed resubmission strategy, method-level technical notes, staffing and milestone plans, and reproducibility-oriented workflow documentation. Together these assets demonstrate readiness to execute a rigorous FIRE-MODEL project with strong potential for field-level adoption.

In final submission form, this section will include concise evidence chains from prior outcomes to present feasibility claims: completed infrastructure, demonstrated workflow reliability, publication and dissemination outcomes, and lessons that directly informed this project's design decisions. The purpose is to show continuity of capability while making clear that the proposed research question is substantively new.

This section will also transparently identify remaining capability gaps and how the proposed work

addresses them. Doing so strengthens credibility by avoiding overstatement and framing the project as a targeted response to known limitations rather than a broad aspirational effort without operational grounding.

In the final proposal package, this section will include concrete placeholders for award numbers, principal outputs, trainee outcomes, and prior dissemination results. Those entries will be limited to verifiable facts and aligned with sponsor formatting rules. The narrative purpose will remain consistent: demonstrate readiness through evidence, delimit claims to supported scope, and show why this next-step project is both ambitious and feasible.

Prior support positioning will also highlight methodological lessons that directly shape this proposal's design. In particular, prior efforts showed that strong aggregate model fit can coexist with weak structural fidelity, and that uncertainty reporting often lags behind modeling sophistication. These lessons motivate the proposal's emphasis on structure-aware validation and calibration-centered interpretation.

Where prior work produced reusable infrastructure, this proposal treats those assets as tested foundations rather than speculative dependencies. Event reconstruction pipelines, data-cube workflows, and early benchmark prototypes reduce startup risk and enable immediate progression to hypothesis testing. This continuity supports feasibility without narrowing ambition.

At submission, we will provide a concise matrix linking prior outputs to planned tasks, indicating which capabilities are mature, which require extension, and which are newly proposed. This matrix will help reviewers distinguish demonstrated readiness from forward-looking innovation and will strengthen the credibility of scope and timeline commitments.

This positioning approach is intentionally conservative in claims and explicit in evidence. By pairing demonstrated infrastructure with clearly bounded innovation targets, the proposal presents a realistic execution pathway that remains scientifically ambitious while reducing programmatic risk.

AI Transparency Statement

We include this statement intentionally as part of an effort to practice clear, accountable, and forward-looking transparency in the use of generative AI in scientific research. Our aim is not only to disclose use, but to help define emerging norms for how researchers can engage with these tools in ways that preserve rigor, authorship, and trust. We view this as an opportunity to contribute constructively to an ongoing conversation about the role of AI in scientific workflows.

Generative AI tools were used in a bounded and explicitly supervised role during the development of this proposal. This included use of ChatGPT (OpenAI, GPT-5.3, 2026) for drafting and iterative refinement of text, and Codex-based systems (OpenAI Codex, 2025–2026 releases) for structured editing, code development, and workflow design. Codex was used both through cloud-based environments (Codes Cloud) and through the Codex desktop application to support coordinated multi-agent editing and longer-running tasks. Prism (OpenAI Prism, web-based research writing environment, 2025–2026) was used in its online form to support LaTeX-native drafting, revision, and integration of scientific content.

These AI-assisted workflows were integrated with standard collaborative and reproducible research tools. GitHub (GitHub, Inc.; web interface and GitHub Desktop application, 2025–2026 versions) was used extensively for version control, repository management, and maintenance of a structured prompt and action log documenting how AI tools were used throughout the development process (`https://github.com/CU-ESIIL/on_growth_and_form`). Google Docs (Google LLC; web-based collaborative editor, 2025–2026) was used for real-time co-author editing and feedback. Zotero (Zotero, version 6.x) was used for citation management and was integrated both within Google Docs and AI-assisted drafting environments to ensure that references were consistently tracked, formatted, and attributed.

AI tools were used to assist with organization, editing, restructuring, and improving clarity of written material, as well as to support development of computational and workflow components. They were not used to generate original scientific ideas, select research directions, or make decisions about modeling, methodology, or interpretation. All hypotheses, modeling frameworks, experimental designs, and conclusions were developed and evaluated by the authors. AI-assisted outputs were treated as drafts and were reviewed line-by-line, revised, and validated by the authors to ensure accuracy, consistency with the literature, and alignment with the intended scientific claims.

A structured prompt and action log was maintained within the project repository (see repository link above) to document where and how AI tools contributed to drafting, editing, and restructuring. This log provides a transparent record of AI-assisted interactions and supports reproducibility and accountability in the development process.

Responsibility for all content in this proposal rests fully with the authors. AI tools were used as assistants in writing, organization, and workflow support, not as sources of authority. We believe that transparent, well-scoped, and well-documented use of AI can improve clarity and efficiency while maintaining the core principles of scientific rigor and authorship. We include this statement not as a disclaimer, but as part of an effort to make AI-assisted scientific workflows more transparent, auditable, and reproducible.

Required Companion Documents (Outside this narrative)

- References Cited
- Data Management and Sharing Plan
- Postdoctoral Mentoring Plan (if applicable)
- Biographical Sketches (SciENcv)
- Current and Pending Support (SciENcv)
- Facilities, Equipment, and Other Resources

Word-Count Target Summary

- Project Summary: Overview — 300

- Project Summary: Intellectual Merit — 225
- Project Summary: Broader Impacts — 225
- Introduction/Motivation — 800
- Hypothesis + Aims — 900
- Data + Measurements — 1,100
- Modeling Framework — 1,300
- V&V + UQ — 1,200
- Work Plan + Deliverables — 1,000
- Team + Convergence — 700
- Broader Impacts Plan — 700
- Prior NSF + Positioning — 500
- **Total target across sections — 8,950**